Five Years Late, Still On Time

mentalhealthonline.org.au
Drupal 8 to 11 journey

DrupalSouth Community Day 2025

About me

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16 years on Drupal.org (alex.skrypnyk)

Drupal and DevOps

Open-source advocate

Founder of DrevOps, a lean digital agency

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About the project

DSCD25

In 2018 **Media Insights** joined forces with **DrevOps**

to build mental health and research platform

for **Swinburne University** in Melbourne

About the project

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- Built in 2018, launched in 2023
- Several additional iterations based on the feedback from closed user groups

More than a CMS

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- **Researcher tools:** therapist/client assignment into control groups; blind and random assignment
- **Therapist–client interaction tools:** private and group messaging
- **Client tools:** interactive exercises, progress tracking and self-assessment
- **Reporting and analytics:** client’s behaviour needs to be tracked for research purposes

The Challenge

Maintaining a **secure**, **feature-complete**, and **production-ready** platform capable of going live **instantly** whenever research required it

💥 Upgrade challenges

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- Custom code needs to be kept up-to-date with the latest PHP version and Drupal core/contribs
- Contrib code changes to handle PHP version upgrades
- Limited budget on updates
- Limited QA resources to test upgrades

🧮 Platform complexity

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🔐 High level of security and user anonymity

💬 Encrypted webforms and messages between users

👥 Role-based permissions with cross-roles workflows

⚙️ Continuous auditing for compliance and stability

⚙️ What's under the hood

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- Drupal 8 → 9 → 10 and now 11
- 150 modules
- Custom theme based on Barrio (Bootstrap)
- PHP 7.4 → 8.3
- Solr search, ClamAV, PDF printing, Highcharts
- Hosted on Acquia Cloud
- Docker, CI and tooling stack based on Vortex

What made continuous upgrades possible

Tooling for continuous upgrades

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🧪 Behat test coverage from the start

⚙️ CI pipelines tested every commit automatically

🧰 Rector for progressive refactoring and modernization

🔄 Renovate to keep dependencies automatically updated

🧪 Behat

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🐘 Written in PHP

No need to learn another language and setup an additional stack

⚡ Fast test preparation via Drupal API

Provisioning 20 articles for a test does not require waiting to click through UI.

🧩 Javascript supported

Interactions requiring Javascript are tested in the same way as non-interactive functionality.

```
@contact @p1
Feature: Contact form

  As anonymous user
  I want to ensure that the contact form is accessible
  In order to be able to contact the site owner

  @api
  Scenario: Anonymous user can use the Contact link and Contact form.
    Given I am an anonymous user
    When I go to "/contact"
    Then I should see the heading Contact
    And I should see "Contact"
    And I should see "Your Name"
    And I should see "Your Email"
    And I should see "Subject"
    And I should see "Message"
    And I should see the button "Send message"

  @api
  Scenario: Anonymous user can fill and submit the contact form
    Given I am an anonymous user
    When I go to "/contact"
    Then I should see the heading Contact
    When I fill in "Name" with "Test User"
    And I fill in "Email" with "test@example.com"
```

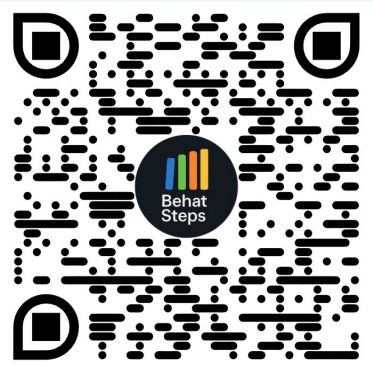
Behat

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Covers critical user journeys:

260 scenarios with 3,000 steps

Behat Steps library was born to reuse the steps across the projects



⚙️ CI pipelines

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Continuous Integration automatically tests, builds, and validates every change. **GitHub Actions** provides:

☁️ Cloud-hosted runners

No setup or servers needed; tests and builds run automatically in GitHub's cloud

🧱 Reusable workflows

Share and standardise CI steps across projects for consistency and speed

🔔 Automatic feedback

Instant test results and build status right in your pull requests

⚙️ CI pipelines

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CI ensured that all automated checks, including Behat and PHPUnit tests, ran automatically and consistently, without manual intervention.

Every change was validated before merging, catching regressions early and keeping the main branch stable.

This automation turned testing into routine practice and made frequent upgrades possible with confidence.

🧰 Rector

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Rector is a CLI tool used for progressive updating of code between PHP and Drupal versions.

⚖️ Rules and rule sets for automated upgrades between language versions

💻 Runs in a command-line as an ad-hoc command by a developer

🤖 Runs in CI to align team coding style and quality

🧰 Rector

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Rector ran 1-2 times a month and automatically updated codebase with the latest changes in the rules.

Updated code ran in CI and was tested with Behat and PHPUnit tests.

Sometimes manual intervention was required when Rector-introduced rules were too prescriptive.

🔄 Renovate

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Renovate is an automated dependency-update tool that scans your codebase for dependencies and opens a pull request if new versions were found.

🪄 Single tool to handle updates across multiple languages

Configurable matchers to support updates of Drupal, PHP, NodeJS dependencies, Docker containers, GitHub actions etc.

📦 Group updates together

Grouping is configurable. Allows to prevent the PR flood.

🗓️ Runs on schedule

Multiple schedules can be configured per group.

🔄 Renovate

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Renovate put the dependency updates on a full auto-pilot with no human interaction: critical updates were checked for twice a day and contrib updates were updated weekly.

Updated code ran in CI and was tested with Behat and PHPUnit tests, then auto-merged to development branch.

Sometimes manual intervention was necessary to re-roll patches for the updated dependencies.



The future

🔮 The future

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🐳 Moving from Acquia to Lagoon for advanced container-based hosting features

Acquia has been great for managed Drupal hosting, but some parts of the application now require custom functionality using custom containers.

A container-first platform like Lagoon lets us run anything the application needs within the same environment stack.

🔮 The future

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👁️‍🗨️ Visual regression during releases with Diffy

Before each release, the new code is deployed to staging while production remains unchanged.

Diffy (https://diffy.website/) automatically captures screenshots from both environments, compares them, and highlights any visual differences for manual review, making regression detection simple and infrastructure-free.

🔮 The future

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🧠 AI instructions from Vortex to speed up development

Vortex includes built-in instructions for AI agents, explaining how the project is structured, which technologies it uses, and how to perform routine development tasks.

This enables AI tools to understand the project context and automate repetitive work safely and accurately.



Lessons learned

💡 Lessons learned

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💸 Continuous upgrades cost less than rebuilds

With around **4 hrs per month** for maintenance and updates,
5 years would be an equivalent of **240 hours** of development time

A full rebuild of this scale would require around **1,000 hours** of development plus **500–600 hours** of project management and client involvement

Lessons learned

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Test coverage is long-term insurance

Behaviour-driven tests focus on what users do, not how the code works.

Tools like Behat make this separation possible, keeping tests valid through major upgrades and reducing QA effort.

💡 Lessons learned

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🛟 Continuous Integration creates safety for innovation

Every change is automatically tested and validated, giving developers confidence to experiment without fear of breaking production.

When something fails, feedback is instant — turning mistakes into quick fixes instead of costly rollbacks.

💡 Lessons learned

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🧘 When “next week” turns into “next year.”

Sometimes everything is done right and on time, yet something outside your control hits pause.

Shifts and pauses are just part of a project’s life cycle.

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Think some of this could help your team or clients?

Ping me on Drupal Slack or drop me a line at alex@drevops.com

All tooling configurations are available in the Vortex -Drupal project template https://www.vortextemplate.com/

Q&A

Thank you!

How to start

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